mutations in Parkinson's disease.[5] Both Gly2019Ser and Ile2020Thr lie within the highly conserved activation segment of the *LRRK2* kinase domain. We postulate that both these pathogenic substitutions have an activating effect on the kinase activity of the LRRK2 protein, causing gain of function, which is compatible with the dominant mode of disease transmission in our families.[4] Several highly potent and selective protein kinase inhibitors are now in development and could offer great therapeutic potential.

We declare that we have no conflict of interest.

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William Nichols and colleagues,[1] Alessio Di Fonzo and colleagues,[2] and William Gilks and colleagues,[3] report the identification of a Gly2019Ser sequence variation contained in leucine-rich repeat kinase 2 (LRRK2; dardarin). Although this variant seems to be associated with Parkinson's disease in a relatively large proportion of patients, an Ile2020Thr variant, which occurs adjacent to Gly2019Ser, has also been seen to segregate with Parkinsonism.[4] Structural and functional analyses of LRRK2 lend support to the proposed pathogenicity of these variants.

LRRK2 is a large protein with multiple domains including several ankyrin, leucine-rich, and WD40 repeats, a Ras-like small GTPase family domain named Roc, and a non-receptor tyrosine kinase-like MAPKKK-related domain. Gly2019Ser and Ile2020Thr are contained in the well studied kinase activation segment of the MAPKKK-related domain (figure).[5] Phosphorylation of specific residues within this segment regulates kinase activity. In particular, the Gly2019 residue is part of the highly conserved DFG-like motif (DYG in LRRK2) at the N-terminus of the activation segment.

The invariant aspartate of the DFG-like motif chelates a magnesium ion that positions the phosphates for phosphotransfer.[5] This magnesium-binding site forms a loop in the active cleft at the interface between the small and large lobe of the kinase domain. Residues in and around the DFG-like motif make essential contributions to the contact surface of the magnesium-binding loop (in the large lobe) to the functionally relevant C-helix of kinases (in the small lobe).[5] Therefore, mutations of Gly2019 and Ile2020 within LRRK2 presumably affect the proper positioning of the C-helix and its catalytically important aminoacids, possibly impairing kinase activity.

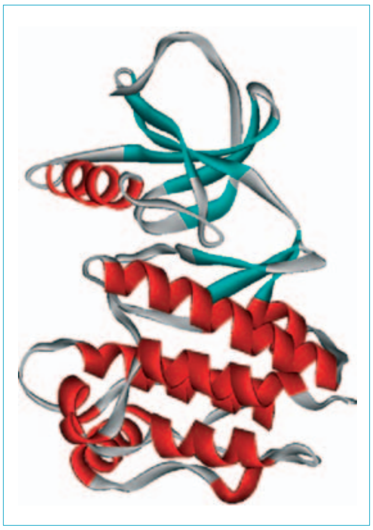

Figure: **Three-dimensional model of LRRK2 based on the B-RAF protein kinase domain**

The additional insights provided into the molecular cause of Parkinson's disease should direct experimental studies and drug design towards the protein kinase activity of LRRK2.

I declare that I have no conflict of interest.

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